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Constraint-Guided BO-GCN Surrogate for Three-Phase Unbalanced Power Flow in DER-Integrated Distribution Systems

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ABSTRACT

Fast and reliable three-phase power flow evaluation is increasingly required in feeders with a high number of distributed energy resources, stochastic photovoltaic generation and electric vehicle charging, where time-series assessment and uncertainty screening may involve thousands of operating points. Classical nonlinear solvers remain reliable but can become computationally intensive when embedded in large scenario loops. Learning surrogates offer speed, yet two gaps persist because vectorised predictors often underuse electrical connectivity. In this paper, an admittance-informed Bayesian-Optimised Graph Convolutional Network model (BO-GCN) is developed to map nodal injections to per-phase voltages and phase angles. Neighbourhood aggregation is weighted by coupling strengths derived from the feeder admittance matrix; a stabilised multi-layer encoder improves training robustness, and a physically structured decoder bounds voltages and enforces a unit-circle sine–cosine angle representation to prevent wrap-around errors. Moreover, OpenDSS-based datasets with 10,000 scenarios are constructed for the IEEE 13-bus and IEEE 123-bus feeders. On the IEEE 13-bus benchmark, relative to a standard GCN, Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) decrease by 31.6% and 26.0%, respectively, and the P95 tail metric drops by 44.3%. Angle MAE and RMSE are reduced by 90.2% and 91.9%, with larger margins over compared models.

1 | Introduction

Modern medium- and low-voltage distribution networks are rapidly evolving from passive, demand-driven infrastructures into active systems with widespread inverter-interfaced Distributed Energy Resources (DER), highly variable Photovoltaic (PV) injections, and flexible Electric Vehicle (EV) charging [1]. In such feeders, phase imbalance is common, and simplifying assumptions such as phase balance or single-phase equivalents can introduce systematic errors when applied across diverse operating points [2]. At the same time, planning and operational workflows increasingly rely on evaluating thousands of scenarios, which turns repeated three-phase unbalanced power flow (PF) calculations into a critical computational burden [3]. A practical way to address this challenge is to replace repeated nonlinear solvers with fast, topology-aware surrogate models that exploit the feeder's graph structure and incorporate physics-guided constraints to preserve consistency and limit violations near operational boundaries [4, 5]. However, existing studies still face important practical limitations related to computational scalability, model robustness under wide operating variations, and the physical consistency of surrogate predictions. Moreover, recent research has highlighted integrated energy systems with cross-energy conversion devices such as gas-fired generation

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and power-to-gas, where interactions between electricity and gas networks make steady-state flow analysis more tightly coupled and model-dependent [6].

Classical models for three-phase unbalanced distribution analysis have focused on improving both formulation robustness and component practicality [7]. A common direction is refining the PF equations using current-injection–type formulations that better handle strong phase asymmetry and practical device models [8]. In parallel, sparse-tableau and node–branch implementations exploit feeder sparsity to reduce computational cost in larger networks while preserving the full three-phase coupling structure [9]. Accurate results also depend on explicit modelling of distribution equipment because these elements directly shape phase-to-phase coupling and constraint activation in real feeders [10]. Load and DER representations further matter in active networks, where voltage dependence and inverter limits can significantly affect phase voltages and conductor loading across operating conditions [11].

Building on these classical models, unbalanced AC optimal PF (ACOPF) extends the same three-phase representation to constrained optimisation [12]. Direct nonlinear unbalanced ACOPF remains the fidelity benchmark but can be computationally heavy when repeated across many operating points [13]. Convex-relaxation approaches improve tractability in some cases, yet their tightness is not guaranteed under tight voltage and current limits and inverter constraints typical of active low-voltage feeders [14, 15]. As distribution operation becomes increasingly scenario-driven, repeated unbalanced ACOPF evaluation across large time series and uncertainty realisations becomes a major scalability limitation [16]. A further practical issue is that many of these classical formulations require well-calibrated three-phase parameters and control settings to remain accurate under wide operating ranges [17]. In distribution feeders, uncertainty in line data, transformer and regulator settings, and load composition can degrade model calibration and increase sensitivity when the same solver is executed repeatedly across diverse scenarios [18]. Consequently, beyond raw computational cost, robustness to parameter and operating-point variability becomes an additional motivation for surrogate models that can learn stable input–output mappings from scenario-rich simulations while still reflecting feeder structure and limits [19]. This has motivated learning-based surrogates for fast prediction [20], but purely data-driven models can be brittle without topology awareness and constraint-consistent training, motivating graph-based and constraint-guided learning directions in recent work [21]. Although classical three-phase unbalanced PF and ACOPF formulations remain the fidelity benchmark, existing studies are still limited by high computational cost in large scenario-based analysis, dependence on well-calibrated network and device parameters, and sensitivity to parameter uncertainty across diverse operating conditions.

Machine Learning (ML) methods have been increasingly adopted to accelerate three-phase unbalanced PF analysis by learning a direct mapping from operating conditions to network states [22]. Early approaches use conventional deep neural networks to predict phase voltages, currents, and losses from inputs such as nodal injections and network parameters, achieving substantial speedups [23]. Similar ideas have been extended toward learning ACOPF-related mappings, where the model predicts either optimal setpoints or key outputs of the optimisation problem to avoid repeated nonlinear optimisation during operation and planning loops [24]. These data-driven surrogates are particularly attractive when thousands of scenarios must be evaluated for time-series assessment and uncertainty analysis [25].

Despite these advances, existing studies still exhibit several practical limitations. In distribution systems, feeders are inherently graph-structured, so vectorised predictors that ignore topology and electrical connectivity can suffer from limited transferability across feeders, reconfiguration states, or even wide operating-range shifts within the same feeder [26]. This has motivated topology-aware learning, enabling message passing aligned with physical connectivity [27]. At the same time, many ML surrogates may produce outputs that appear numerically plausible but violate PF consistency or operational limits when extrapolating beyond the training distribution, particularly near voltage and thermal boundaries where reliability matters most [28]. Accordingly, recent research increasingly emphasises constraint-guided learning to improve physical consistency and reduce unsafe predictions under diverse operating conditions [29]. Recent topology-aware surrogates therefore rely on graph neural networks that propagate information along feeder connectivity while injecting electrical context through edge features so that aggregation reflects coupling strength rather than adjacency alone [30]. In parallel, constraint-guided learning has been implemented not only as generic penalty terms but also through structured representations that improve numerical behaviour, such as handling angle periodicity and discouraging infeasible voltage predictions, so that accuracy remains reliable near operating limits where violations are most consequential [31]. Moreover, recent studies on integrated energy systems have shown that cross-energy conversion devices such as gas-fired generators and power-to-gas units introduce bidirectional interdependencies between electricity and gas networks and require additional coupled state variables beyond conventional electrical PF formulations [32]. Therefore, despite recent progress, there is still a practical need for designs that jointly encode electrically meaningful coupling in message passing and stabilise deep graph encoders to generalise across wide DER- and EV-driven operating ranges, while recognising that the present study focuses on single-carrier three-phase electrical distribution PF.

Motivated by these gaps, the main contributions of this paper are as follows:

- A BO-GCN surrogate architecture is developed specifically for three-phase unbalanced PF prediction, where neighbourhood aggregation is informed by coupling strengths derived from the feeder admittance matrix so that information exchange reflects the actual electrical structure of the network.
- A physics-guided voltage prediction mechanism is introduced by combining bounded magnitude estimation with a circular sine–cosine angle representation and unit-circle normalisation, thereby improving feasibility, numerical stability, and angular consistency.
- A stabilised deep graph-convolution encoder is designed through the integration of stacked graph-convolution layers, per-layer normalisation, and dropout, enabling more robust

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learning of the nonlinear power-flow mapping under diverse operating conditions.

- A compact two-layer nonlinear decoder is employed to map node embeddings to voltage states, yielding a more expressive output transformation than a single linear head while maintaining a lightweight and computationally efficient surrogate model.

2 | Unbalanced Distribution Network Model

In unbalanced distribution networks, phase coupling and the coexistence of single, two, and three-phase lines must be represented explicitly. For each line $(i, j) \in \mathcal{M}$, let $\Gamma_{ij} \subseteq \{a, b, c\}$ denote the set of available phases on that line. The set $\Gamma_i \subseteq \Gamma$ identifies the phases physically present at bus i. The set $\Gamma_{ij} \subseteq \Gamma$ identifies the phases present on branch (i, j), and only those bus–phase nodes (i, ϕ) with $\phi \in \Gamma_i$ are included in $\mathcal{K}$. A one-to-one indexing map $\pi(\cdot)$ assigns each bus–phase pair $(i, \phi) \in \mathcal{K}$ to a unique node index $\kappa = \pi(i, \phi) \in \{1, \ldots, n\}$, and the stacked bus–phase voltage and current vectors $V \in \mathbb{C}^n$ and $I \in \mathbb{C}^n$ are formed according to this ordering. The phase-voltage phasors at buses i and j are collected in the vectors $V_i = [V_i^{\phi}]_{\phi \in \Gamma_{ij}}$ and $V_j = [V_j^{\phi}]_{\phi \in \Gamma_{ij}}$, while the corresponding line-current vector is $I_{ij} = [I_{ij}^{\phi}]_{\phi \in \Gamma_{ij}}$. The electrical characteristics of each line are captured by the phase-coupled admittance matrix $Y_{ij} \in \mathbb{C}^{|\Gamma_{ij}| \times |\Gamma_{ij}|}$, whose off-diagonal entries model mutual coupling. The voltage–current relationship on line (i, j) can then be written as Equation (1).

$$\mathrm{I}_{ij} = \mathrm{Y}_{ij} \ \left(\mathrm{V}_i - \mathrm{V}_j\right) \ \forall (i, j) \in \mathcal{M} \tag{1}$$

When $|\Gamma_{ij}| = 3$, Equation (1) can be expanded, which explicitly shows that each phase current depends not only on its own voltage drop but also on the voltage drops of the other phases through the mutual admittances.

$$\begin{bmatrix} I_{ij}^a \\ I_{ij}^b \\ I_{ij}^c \end{bmatrix} = \begin{bmatrix} Y_{ij}^{aa} & Y_{ij}^{ab} & Y_{ij}^{ac} \\ Y_{ij}^{ba} & Y_{ij}^{bb} & Y_{ij}^{bc} \\ Y_{ij}^{ca} & Y_{ij}^{cb} & Y_{ij}^{cc} \end{bmatrix} \begin{bmatrix} V_i^a - V_j^a \\ V_i^b - V_j^b \\ V_i^c - V_j^c \end{bmatrix} \tag{2}$$

Accordingly, the current of phase $\varphi \in \Gamma_{ij}$ can be expressed as Equation (3).

$$I_{ij}^{\varphi} = \sum_{\psi \in \Gamma_{ij}} Y_{ij}^{\varphi\psi} \ \left(V_i^{\psi} - V_j^{\psi}\right) \ \forall (i, j) \in \mathcal{M}, \ \forall \varphi \in \Gamma_{ij} \tag{3}$$

Based on the per-phase current, the sending-end complex PF on phase φ is defined by Equation (4),

$$S_{ij}^{\varphi} = V_i^{\varphi} \ (I_{ij}^{\varphi})^* \ \forall (i, j) \in \mathcal{M}, \ \forall \varphi \in \Gamma_{ij} \tag{4}$$

where $(.)^*$ denotes complex conjugation. By stacking the per-phase powers, the line complex power vector can be given in Equation (5).

$$S_{ij} = diag\,(V_i)\ I_{ij}^* \tag{5}$$

Finally, substituting Equation (3) into Equation (4) yields,

$$S_{ij}^{\varphi} = V_i^{\varphi} \left(\sum_{\psi \in \Gamma_{ij}} Y_{ij}^{\varphi\psi} \left(V_i^{\psi} - V_j^{\psi}\right) \right)^* \ \forall (i, j) \in \mathcal{M}, \ \forall \varphi \in \Gamma_{ij} \tag{6}$$

which highlights that, due to the off-diagonal coupling terms $Y_{ij}^{\varphi\psi}$ for $\varphi \neq \psi$, the complex power on a given phase depends on voltage differences across all phases present on the line.

Followed by Equation (6), the unbalanced PF model is closed by enforcing nodal current balance over the bus-phase network, which can be written as Equation (7).

$$I_{inj} = YV \tag{7}$$

where $Y \in \mathbb{C}^{n \times n}$ is the bus–phase nodal admittance matrix assembled over $\mathcal{K}$. For a constant PQ injection at node $\kappa \in \{1, \ldots, n\}$ under scenario s, with $S_{\kappa}^{(s)} = P_{\kappa}^{(s)} + jQ_{\kappa}^{(s)}$, the current injection is given in Equation (8).

$$I_{inj,\kappa}^{(s)} = \left(\frac{S_{\kappa}^{(s)}}{V_{\kappa}^{(s)}} \right)^* \tag{8}$$

These equations complete the three-phase unbalanced power-flow model used for collecting reference solutions.

3 | Proposed Methodology

3.1 | Proposed Model Formulation

Unlike generic surrogate architectures that treat the PF mapping mainly as a high-dimensional input-output regression task, the proposed BO-GCN is structured to reflect the electrical and operational characteristics of three-phase unbalanced distribution feeders. The architecture is therefore designed so that the graph propagation stage captures admittance-mediated bus–phase interactions, while the output stage enforces physically meaningful voltage magnitude and angle representations. In this way, the proposed model is not only optimised for predictive accuracy, but also tailored to the phase-coupled and constraint-sensitive nature of unbalanced PF behaviour.

The proposed architecture combines both adapted and problem-specific elements. Utilising graph convolution layers, layer normalisation, dropout, and an MLP-based output head follows established deep-learning practice and is adopted here to improve optimisation stability and representational capacity. Likewise, Bayesian Optimisation (BO) is used as a systematic tuning strategy rather than as a standalone methodological contribution. The contribution of the present work lies in specialising these components to the bus-phase unbalanced PF setting through admittance-informed graph construction, electrically weighted propagation, and physically constrained voltage-angle decoding.

The proposed BO-GCN learns the steady-state voltage profile of an unbalanced distribution network by mapping scenario-wise constant-PQ injections at bus–phase nodes to bus–phase voltage magnitude and angle. The electrical network is phase-

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coupled and consistent with the unbalanced branch model in Equation (1–6), where each line $(i, j) \in \mathcal{M}$ has an available phase set $\Gamma_{ij} \subseteq \{a, b, c\}$ and phase couplings are captured by the complex admittance entries. Leveraging this physical structure, the learning model is constructed directly from the bus–phase nodal admittance matrix.

Let the bus-phase representation of the feeder be encoded as a directed graph $\mathcal{G} = (\mathcal{K}, \mathcal{E})$, where each node $\kappa \in \mathcal{K}$ corresponds to one bus-phase. The bus-phase nodal admittance matrix is denoted by $\mathrm{Y} \in \mathbb{C}^{|\mathcal{K}|\times|\mathcal{K}|}$. An edge is created whenever the corresponding off-diagonal coupling is nonzero.

$$(\kappa, \mu) \in \mathcal{E} \Leftrightarrow |\, Y_{\kappa\mu} \,| > \varepsilon \quad \kappa \neq \mu \tag{9}$$

For each edge (κ, μ), the real and imaginary parts of $Y_{\kappa\mu}$ are stored as edge attributes and a scalar coupling weight is defined as the admittance magnitude, shown in Equations (10) and (11), respectively [33].

$$\mathrm{e}_{\kappa\mu} = \begin{bmatrix} G_{\kappa\mu} \\ B_{\kappa\mu} \end{bmatrix} \quad Y_{\kappa\mu} = G_{\kappa\mu} + jB_{\kappa\mu} \tag{10}$$

$$w_{\kappa\mu} = \sqrt{G_{\kappa\mu}^2 + B_{\kappa\mu}^2} = |\, Y_{\kappa\mu} \,| \tag{11}$$

This construction ensures that message passing follows the same coupling pattern that governs the unbalanced network equations, rather than using a purely topological adjacency.

For each operating scenario $s \in \{1, \ldots, N_s\}$, the node input feature is the constant PQ net injection at node κ,

$$\tilde{P}_\kappa^{(s)} = \frac{P_\kappa^{(s)} - \mu_P}{\sigma_P} \tag{12}$$

$$\tilde{Q}_\kappa^{(s)} = \frac{Q_\kappa^{(s)} - \mu_Q}{\sigma_Q} \tag{13}$$

$$\mathrm{x}_\kappa^{(s)} = \begin{bmatrix} \tilde{P}_\kappa^{(s)} \\ \tilde{Q}_\kappa^{(s)} \end{bmatrix} \tag{14}$$

where μ_P, σ_P and μ_Q, σ_Q are calculated from the training set to standardise the inputs. The learning target is the bus-phase voltage in magnitude–angle form, $\mathrm{y}_\kappa^{(s)} = [\, |V_\kappa^{(s)}| \;\; \sin\theta_\kappa^{(s)} \;\; \cos\theta_\kappa^{(s)} \,]^T$ with a sine–cosine embedding to avoid 2π discontinuities.

Moreover, the node features and targets are collected into scenario-level matrices, enabling a matrix-form representation of the subsequent graph convolution operations. Let $H^{(0,s)} \in \mathbb{R}^{|\mathcal{K}|\times 2}$ denote the stacked node features for scenario s, expressed as Equation (15).

$$H^{(0,s)} = [x_\kappa^{(s)}]_{\kappa\in\mathcal{K}} \tag{15}$$

Using the edge weights $w_{\kappa\mu}$, the weighted adjacency and degree matrices are defined by Equation (16).

$$[W]_{\kappa\mu} = \begin{cases} w_{\kappa\mu} & (\kappa, \mu) \in \mathcal{E} \\ 0 & \text{otherwise} \end{cases} \rightarrow D = diag\left(\sum_\mu [W]_{\kappa\mu}\right)_{\kappa\in\mathcal{K}} \tag{16}$$

Self-loops are added to stabilise propagation, and the normalised propagation operator is shown in Equation (17).

$$S = \tilde{D}^{-1/2}\, \tilde{W}\, \tilde{D}^{-1/2} \tag{17}$$

Here, $\tilde{W} = W + I$ adds self-loops and $\tilde{D} = diag(\sum_\mu [\tilde{W}]_{\kappa\mu})$ is the corresponding degree matrix, while S is the symmetrically normalised propagation operator that stabilises and balances message passing across nodes.

For L graph-convolution layers, the encoder updates are expressed as Equation (18),

$$H^{(\ell+1,s)} = Drop\left(\sigma\left(LN\left(SH^{(\ell,s)}\Theta^{(\ell)}\right)\right)\right) \;\; \ell = 0, \ldots, L-1 \tag{18}$$

where $\Theta^{(\ell)}$ are trainable weights, $LN(\cdot)$ is layer normalisation, $\sigma(\cdot)$ is Rectified Linear Unit (ReLU), and $Drop(\cdot)$ is dropout. This design emphasises electrically strong couplings during aggregation and suppresses weak interactions, thereby embedding a physics-aligned inductive bias.

A node-wise Multi-Layer Perceptron (MLP) maps the final embedding $h_\kappa^{(s)}$ to raw outputs $z_\kappa^{(s)} = [z_1, z_2, z_3]^T$.

Furthermore, voltage magnitude is constrained to $[V_{min}, V_{max}]$ through a sigmoid mapping, and angle is predicted in the sine–cosine space using bounded nonlinearities, both of which are shown in Equations (19) and (20), respectively.

$$\widehat{|V|}_\kappa^{(s)} = V_{min} + (V_{max} - V_{min})\; sigmoid\,(z_1) \tag{19}$$

$$\tilde{s}_\kappa^{(s)} = tanh\,(z_2) \;\; \tilde{c}_\kappa^{(s)} = tanh\,(z_3) \tag{20}$$

Since $\tilde{s}_\kappa^{(s)}, \tilde{c}_\kappa^{(s)}$ may violate the unit-circle constraint, they are explicitly normalised to obtain $\hat{s}_\kappa^{(s)}$ and $\hat{c}_\kappa^{(s)}$ such that $(\hat{s}_\kappa^{(s)})^2 + (\hat{c}_\kappa^{(s)})^2 = 1$, where ϵ is a small constant added for numerical stability.

$$\hat{s}_\kappa^{(s)} = \frac{\tilde{s}_\kappa^{(s)}}{\sqrt{\left(\tilde{s}_\kappa^{(s)}\right)^2 + \left(\tilde{c}_\kappa^{(s)}\right)^2 + \epsilon}} \hat{c}_\kappa^{(s)} = \frac{\tilde{c}_\kappa^{(s)}}{\sqrt{\left(\tilde{s}_\kappa^{(s)}\right)^2 + \left(\tilde{c}_\kappa^{(s)}\right)^2 + \epsilon}} \tag{21}$$

Therefore, the final prediction at node κ is written as $\hat{\mathrm{y}}_\kappa^{(s)} = [\, \widehat{|V|}_\kappa^{(s)} \;\; \hat{s}_\kappa^{(s)} \;\; \hat{c}_\kappa^{(s)} \,]^T$.

With the forward mapping from bus–phase injections to voltage outputs established, the model parameters are learned by minimising a supervised objective that jointly penalises voltage-magnitude and angle prediction errors across all bus–phase nodes. Specifically, training minimises a composite loss comprising a Smooth-ℓ_1 term for voltage magnitude, parameterised by $\beta > 0$, and an angle term defined in the sine–cosine space, both of which can be given in Equations (22) and (23), respectively.

$$\mathcal{L}_V = \frac{1}{|\,\mathcal{K}\,|} \sum_{\kappa\in\mathcal{K}} Huber_\beta\left(\widehat{|V|}_\kappa^{(s)} \, |V|_\kappa^{(s)}\right) \tag{22}$$

$$\mathcal{L}_\theta = \frac{1}{|\,\mathcal{K}\,|} \sum_{\kappa\in\mathcal{K}} \| \begin{bmatrix} \hat{s}_\kappa^{(s)} \\ \hat{c}_\kappa^{(s)} \end{bmatrix} - \begin{bmatrix} \sin\theta_\kappa^{(s)} \\ \cos\theta_\kappa^{(s)} \end{bmatrix} \|_2^2 \tag{23}$$

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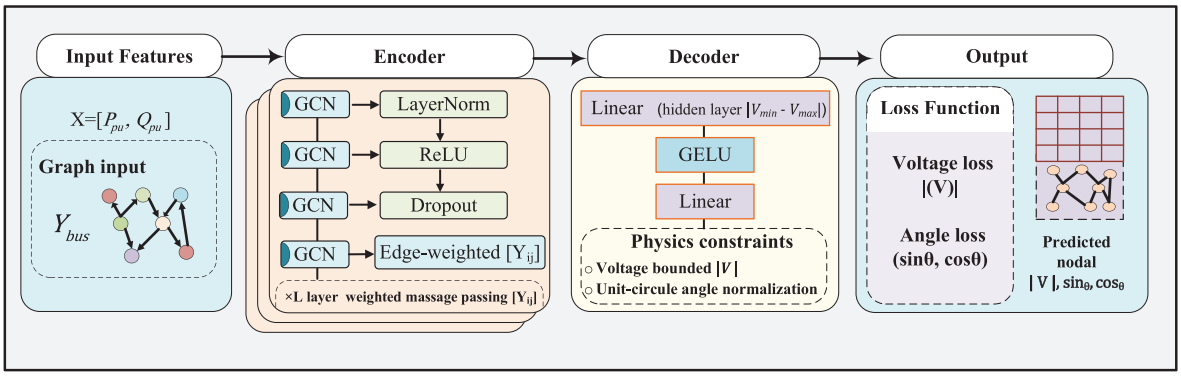


FIGURE 1 | Architecture of the proposed BO-GCN model for three-phase unbalanced PF.

Therefore, the total scenario loss is expressed as Equation (24), and the empirical risk over the training set S_{tr} is minimised, given in Equation (25).

$$\mathcal{L}^{(s)}(\Theta) = \lambda_V \mathcal{L}_V + \lambda_\theta \mathcal{L}_\theta \tag{24}$$

$$\min_{\Theta} \frac{1}{|S_{tr}|} \sum_{s \in S_{tr}} \mathcal{L}^{(s)}(\Theta) \tag{25}$$

Figure 1 illustrates the architecture of the proposed BO-GCN surrogate for three-phase unbalanced PF prediction. The model is designed to address two practical issues observed in purely data-driven surrogates, namely, the underuse of electrical connectivity and the risk of producing numerically plausible voltages that violate basic consistency or operating limits.

Accordingly, the input to the network consists of node features $X = [P_{pu}, Q_{pu}]$ and an electrical graph derived from the feeder admittance matrix Y_{bus}, so information is exchanged along physically coupled bus–phase nodes rather than through an arbitrary vector mapping. In the encoder, multiple graph-convolution layers perform message passing with edge weights computed from admittance couplings $|Y_{ij}|$, which prioritises stronger electrical interactions during neighbourhood aggregation. Layer normalization, a ReLU nonlinearity, and dropout are applied across stacked GCN layers to stabilise optimisation and improve generalisation under wide operating-condition shifts. The decoder then maps the learned node embeddings to voltage states using a compact two-layer MLP with a Gaussian Error Linear Unit (GELU) activation, providing a richer nonlinear output head than a single linear layer while keeping inference lightweight. Physical structure is enforced at the output level by bounding the predicted voltage magnitudes within $[V_{\min}, V_{\max}]$ and representing angles via a $(\sin\theta, \cos\theta)$ pair followed by unit-circle normalisation to prevent wrap-around errors and improve angular consistency. Training is driven by a composite objective that penalises voltage-magnitude error and angle error in the sine–cosine space, aligning the learned mapping with both accuracy and physically meaningful output geometry.

3.2 | Hyperparameter Tuning

To avoid ad hoc tuning, BO searches over a hyperparameter space $\mathcal{H}$. Here, BO is adopted as a standard yet systematic model-selection strategy to avoid ad hoc hyperparameter tuning. The methodological novelty of the study does not lie in BO itself, but in the feeder-aware BO-GCN formulation that is being tuned. Let η represent the hidden width, number of GCN layers, dropout rate, Learning Rate (LR), weight decay, batch size, loss weights, Huber parameter, and voltage bounds. The BO objective evaluates the validation risk after T_{bo} training epochs, after which the final model is trained using η^* with early stopping.

$$\eta^* = \arg\min_{\eta \in \mathcal{H}} \mathcal{J}(\eta) = \frac{1}{|S_{va}|} \sum_{s \in S_{va}} \mathcal{L}^{(s)}\left(\Theta_{T_{bo}}(\eta)\right) \tag{26}$$

BO is performed only as an offline hyperparameter-selection stage during model development. After the best configuration η^* is identified, the final BO-GCN is retrained once using the selected setting and then deployed with fixed hyperparameters. Therefore, the practical computational burden of BO is associated with offline model development rather than online inference. For very large distribution systems, this offline cost may increase because each BO trial requires training and validation of a candidate configuration, and the cost per trial grows with graph size, edge count and architectural dimensions. Accordingly, the proposed approach is most practical when the BO search space remains compact and physically motivated and when retuning is performed only if the network model or operating regime changes significantly. Furthermore, beyond pure prediction error, the method quantifies physical consistency using a KCL mismatch calculated from Y. First, the predicted complex voltage phasor is reconstructed from the constrained outputs $\hat{V}_\kappa^{(s)} = \widehat{|V|}_\kappa^{(s)} (\hat{c}_\kappa^{(s)} + j\,\hat{s}_\kappa^{(s)})$. Stacking all nodes yields $\hat{V}^{(s)} = [\hat{V}_\kappa^{(s)}]_{\kappa \in \mathcal{K}}$. Here, the current injection is reconstructed from the full predicted complex voltage phasor, not from the voltage magnitude alone. Specifically, after forming $\hat{V}_\kappa^{(s)}$, the implied current at each constant-PQ bus–phase node is computed through the same algebraic power-current relation used in the reference unbalanced PF model. Given constant PQ injections $S_\kappa^{(s)} = P_\kappa^{(s)} + jQ_\kappa^{(s)}$, the implied injection current and the nodal KCL residual are then shown as Equations (27) and (28), respectively.

$$\hat{I}_{\kappa,inj}^{(s)} = \left(\frac{S_\kappa^{(s)}}{\hat{V}_\kappa^{(s)}}\right)^* \quad \hat{I}_{inj}^{(s)} = [\hat{I}_{\kappa,inj}^{(s)}]_{\kappa \in \mathcal{K}} \tag{27}$$

$$r_{KCL}^{(s)} = Y\,\hat{V}^{(s)} - \hat{I}_{inj}^{(s)} \tag{28}$$

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TABLE 1 | Optimal BO-GCN hyper parameter configuration selected by BO.

Parameter	Hidden layers	Num. GCN	Dropout	LR	Weight decay	Batch size
Value	128	3	0.1122	0.0012	1.1×10^{-8}	8

A normalised per-node residual magnitude is then defined by Equation (29).

$$\rho_{\kappa}^{(s)} = \frac{\mid r_{KCL,\kappa}^{(s)} \mid}{\sum_{\mu \in \mathcal{K}} \mid Y_{\kappa\mu} \mid + \epsilon} \tag{29}$$

This diagnostic is intentionally phasor-sensitive. Because $\hat{I}_{\kappa,inj}^{(s)}$ is calculated from the full complex voltage phasor, both voltage-magnitude and angle errors affect the resulting KCL residual. In particular, if the predicted voltage is written as $\hat{V}_{\kappa} = |V_{\kappa}|\, e^{j(\theta_{\kappa}+\delta_{\kappa})}$ while the magnitude is accurate and only the angle is perturbed by a small error δ_{κ}, then the reconstructed current becomes $\hat{I}_{\kappa,inj} = I_{\kappa,inj}\, e^{j\delta_{\kappa}}$, so that the current-injection error scales approximately with $|I_{\kappa,inj}||\delta_{\kappa}|$ for small δ_{κ}. This sensitivity is desirable, since current-balance consistency in unbalanced PF depends on the full voltage phasor rather than on voltage magnitude alone. At the same time, the bounded decoding in Equation (19) keeps $\hat{V}_{\kappa}$ within $[V_{min}, V_{max}]$, which avoids uncontrolled amplification in the reconstructed current due to very small voltage magnitudes. Therefore, the KCL residual is used as a complementary physics-orientated diagnostic that jointly reflects phasor consistency with the network equations. These quantities provide a physics-aligned diagnostic, expressing if predicted voltages are consistent with the network coupling encoded by Y and the injected PQ; the residuals remain small. When uncertainty estimates are required, dropout is activated during inference to generate M stochastic forward passes, from which a quantile-based prediction interval for voltage magnitude is obtained through Equation (30).

$$PI_{\kappa}^{(s)}(\alpha) = [\, Q_{(1-\alpha)/2}(\{|\widehat{V}|_{\kappa}^{(s,m)}\}_{m=1}^{M}) Q_{(1+\alpha)/2}(\{|\widehat{V}|_{\kappa}^{(s,m)}\}_{m=1}^{M}) \,] \tag{30}$$

This enables reporting empirical coverage and interval width across the operating space without modifying the core deterministic predictor.

Table 1 summarises the optimal hyperparameters selected by BO for the BO-GCN model on the IEEE 123-bus system. Hyperparameter selection is performed by minimising the mean validation value of the composite training loss, defined as a weighted sum of a Huber loss on voltage magnitudes and a mean-squared error on the sine–cosine representation of voltage angles, with weights w_V and w_{Ang}. The search space covers the hidden dimension, the number of GCN layers, dropout rate, learning rate, weight decay, and batch size. The resulting best trial is fixed and used in all subsequent experiments to ensure reproducibility and enable fair comparisons across models.

3.3 | Evaluation Performance Metrics

Model performance is quantified using three complementary error metrics calculated over the test set, namely, the Mean Absolute Error (MAE), the Root Mean Squared Error (RMSE), and the 95th-percentile absolute error (P95). These metrics are reported for the predicted voltage states and are evaluated across all bus–phase nodes and all test scenarios.

Let y_t and $\hat{y}_t$ denote the ground-truth and predicted values of a given target component at sample index t, where the index t enumerates all bus–phase nodes over all scenarios in the evaluation set. Therefore, the MAE is defined as Equation (31),

$$\text{MAE} = \frac{1}{N} \sum_{t=1}^{N} \mid y_t - \hat{y}_t \mid \tag{31}$$

where N is the total number of evaluated samples. MAE provides an interpretable average deviation and is less sensitive to a small number of large errors.

To emphasise larger deviations, RMSE is calculated by Equation (32).

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{t=1}^{N} (y_t \hat{y}_t)^2} \tag{32}$$

Because the squared term penalises large errors more strongly, RMSE is particularly informative for identifying cases where the surrogate produces occasional but significant mispredictions.

Moreover, tail performance is assessed using the 95th-percentile absolute error (P95). Let $e_t = |y_t - \hat{y}_t|$ denote the absolute error of sample t, and collect all absolute errors over the evaluation set into the sequence $(e_1, e_2, \ldots, e_N)$. The P95 is defined as the 0.95-quantile of this sequence, denoted by $Q_{0.95}$, shown in Equation (33).

$$P95 = Q_{0.95}\ (e_1, e_2, \ldots, e_N) \tag{33}$$

Equivalently, letting $(e_{(1)} \le e_{(2)} \le \ldots \le e_{(N)})$ denote the non-decreasingly sorted absolute errors, a common empirical definition is $\text{P95} = e_{(\lceil 0.95\, N \rceil)}$, so that approximately 95% of samples satisfy $e_t \le \text{P95}$. Percentiles beyond P95 are reported analogously by replacing 0.95 with the desired percentile level in Equation (33).

Finally, Figure 2 summarises the overall workflow of the study, from OpenDSS-based scenario construction and graph dataset preparation to BO-GCN model training and performance evaluation.

4 | Results

In this section, we assess the proposed BO-GCN via extensive simulations on three standard unbalanced distribution feeders. The IEEE 13-bus system [34] is used as the primary benchmark to evaluate how accurately the model captures phase-coupled voltage behaviour under wide variations of load, PV output and EV charging. The single-line diagram of the modified IEEE 13-bus feeder is shown in Figure 3.

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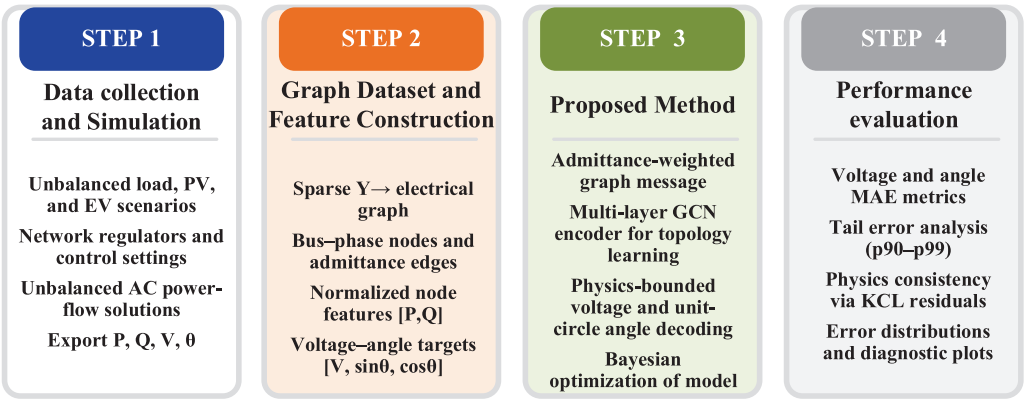


FIGURE 2 | Overall workflow of the proposed approach for three-phase unbalanced distribution PF.

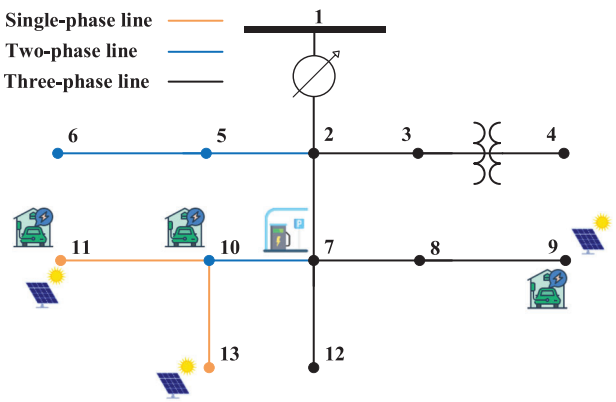


FIGURE 3 | Single-line diagram of the modified IEEE 13-bus unbalanced distribution system.

Scalability and robustness are then examined on the IEEE 123-bus test feeder, which features pronounced unbalance and multi-phase line configurations. Furthermore, scenario-based datasets are collected from OpenDSS [35] using 10,000 operating points with stochastic load scaling, irradiance-driven PV variability, and stochastic EV charging and discharging. More specifically, the scenario set was designed to span a broad operating envelope relevant to safe feeder operation. In the collected dataset, PV uncertainty is represented through scenario-dependent irradiance scaling with small unit-level perturbations, while EV uncertainty is represented through stochastic charging-discharging activation, randomly varying active-power commands, and voltage-dependent Volt–Var response. In addition, load demand is varied under both normal and extreme conditions so that the learned surrogate is exposed to a wide range of realistic network states rather than a narrow deterministic operating point. Per-phase nodal voltages and branch currents are recorded as targets, while nodal injections and network topology information form the inputs.

In the OpenDSS reference condition, the total three-phase demand is approximately 3.45 MW and 2.09 MVAr, with phase totals of $P_A \approx 1.21$ MW, $P_B \approx 0.98$ MW, and $P_C \approx 1.27$ MW. The per-phase base power is 5/3 MVA, the nominal primary line-to-line voltage is 4.16 kV, and 2.401 kV is used for single-phase branches. For this study, a modified IEEE 13-bus feeder is considered with three PV units and four EV units placed at

TABLE 2 | The main specifications of PV and EV units in the modified IEEE 13-bus test system.

Unit	Location		Rated power (kW)
	Bus	Phase	
PV	9	abc	345
	11	c	173
	13	a	173
EV	7	abc	200
	9	b	15
	10	a	10
	11	c	20

selected buses. The PV systems operate at unity power factor with a daily irradiance profile, while EVs follow a four-point Volt–Var characteristic with stochastic charging and discharging across scenarios. The main PV and EV parameters are summarised in Table 2.

Furthermore, Figure 4 illustrates the learning curve of the BO-GCN model, showing the evolution of the training and validation

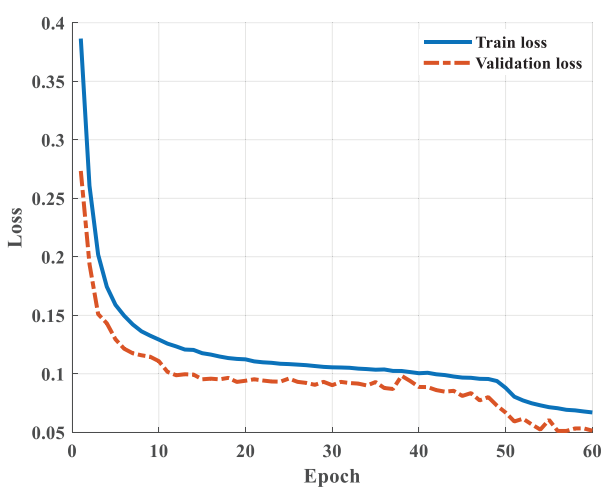


FIGURE 4 | Learning curve of the BO-GCN model on the IEEE 123-bus dataset.

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TABLE 3 | Performance comparison of the evaluated models on the IEEE 13-bus test case.

	MAE		RMSE		P95	
Model	**Voltage (ΔV)**	**Angle ($\Delta\theta$)**	**Voltage (ΔV)**	**Angle ($\Delta\theta$)**	**Voltage ($\lvert\Delta V\rvert$)**	**Angle ($\lvert\Delta\theta\rvert$)**
BO-GCN	0.0039	0.025	0.0094	0.0357	0.0162	0.0569
GCN	0.0057	0.2542	0.0127	0.4381	0.0291	0.6645
GAT	0.0062	0.5392	0.013	0.7221	0.0312	1.3956
TCN	0.0118	0.9609	0.0159	0.9037	0.0328	2.248

losses over epochs. The learning curve demonstrates stable and well-behaved convergence of the BO-GCN training process. Both the training and validation losses drop sharply during the first few epochs, followed by a slower, steady improvement as optimisation refines the solution. The validation loss remains consistently below the training loss throughout training, which is expected since dropout is active during training but disabled during validation, yielding slightly lower validation errors. No divergence or widening gap between the curves is observed, indicating good generalisation without overfitting. The lowest validation loss is achieved near the final epochs, confirming that the selected training horizon is sufficient for convergence.

4.1 | Case 1: IEEE 13-Bus System

In order to provide a clear baseline comparison across the considered surrogates, the IEEE 13-bus feeder results are examined. It focuses on the prediction errors of voltage magnitude and phase angle, reported through MAE, RMSE and the 95th-percentile (P95) absolute error metrics. The corresponding accuracy statistics for all models are summarised in Table 3.

As summarised in Table 3, the consistently lower voltage errors of BO-GCN can be attributed to the topology-aware message passing, so a graph convolution that aggregates neighbour information provides the correct inductive bias. Bayesian hyperparameter tuning then becomes critical because the accuracy and stability tradeoff in GCNs is sensitive to depth, hidden size, normalisation, and regularisation. The reduced P95(|ΔV|) under BO-GCN is particularly meaningful: it suggests that the tuned model is not only accurate on average but also more reliable under stressed operating points, where unbalanced loading amplifies nonlinear sensitivities and produces heavier error tails in less-regularised models. Moreover, for angle estimation, the spread across models is expected to be larger because angles are more sensitive to reference effects and to small mismatches in power-flow coupling. The markedly heavier tails in GCN and GAT indicate that, while they can fit typical operating points, they sometimes incur rare but large-angle errors, consistent with instability in the learned mapping for certain scenarios. BO-GCN mitigates this by selecting proper hyperparameters that better control generalisation, yielding both low average and low high-percentile angle errors. The attention mechanism in GAT does not improve results here, which is plausible for a compact feeder where attention can introduce additional variance and sensitivity without adding essential structural information beyond what the base graph already encodes. Finally, the non-graph TCN can achieve moderate average angle accuracy by exploiting the fixed ordering of bus-phase nodes as an implicit positional prior, but its substantially larger P95 for angle and weaker voltage performance reflect the limitation of ignoring the true electrical connectivity.

While Table 3 summarises the aggregate error statistics, the predicted-actual scatter plots in Figures 5(a)-5(d) show where each model succeeds or fails across the operating range. The diagonal line represents perfect agreement and the color scale represents the local absolute error.

As shown in Figure 5(a), BO-GCN forms the most compact and well-balanced cloud around the $x = y$ line, and it avoids the distinct high-error pockets that appear near the edges of the distribution in the baselines. This behaviour is consistent with Bayesian hyperparameter tuning improving the balance between model complexity and regularisation, so the mapping remains stable when scenarios move toward less frequent operating conditions. The plain GCN in Figure 5(b) preserves the overall trend but shows a wider spread and a clear concentration of larger errors at the upper voltage end, which indicates weaker robustness when the operating conditions deviate from the dominant training mass and weaker control of error growth in the tails. GAT in Figure 5(c) appears visually closer to the $x = y$ line in the dense central region, yet it still produces occasional high-error strands and outliers at high voltage values. This means the bulk alignment can look strong while tail behaviour remains fragile under rare operating combinations. Finally as demonstrated in Figure 5(d), TCN exhibits the broadest dispersion and structured departures from the $x = y$ line, which is expected because a non-graph sequence model cannot represent feeder-dependent sensitivities and phase and topology couplings, so the errors become systematic rather than purely random as scenarios become more challenging.

To assess robustness under rare operating points driven by load variations, variable PV injection, and different initial SOC of EVs, Figures 6(a)-6(b) shifts the focus from the central tendency to the tail behaviour of the absolute angle deviation. The tail ratio plot in Figure 6(a) quantifies how often each model produces large angle deviations compared with the proposed BO-GCN, while the CCDF plot in Figure 6(b) reports the exceedance probability directly as a function of the threshold. For a threshold t in degrees, the tail ratio compares the empirical probability $P(|\Delta\theta| \geq t)$ of a given model against the same probability for BO-GCN. A value of one indicates parity with the proposed method, whereas values above one mean that large deviations occur more frequently than in BO-GCN.

As illustrated in Figure 6(a), Across thresholds, GCN exhibits a consistently elevated tail ratio, showing that moderate to large

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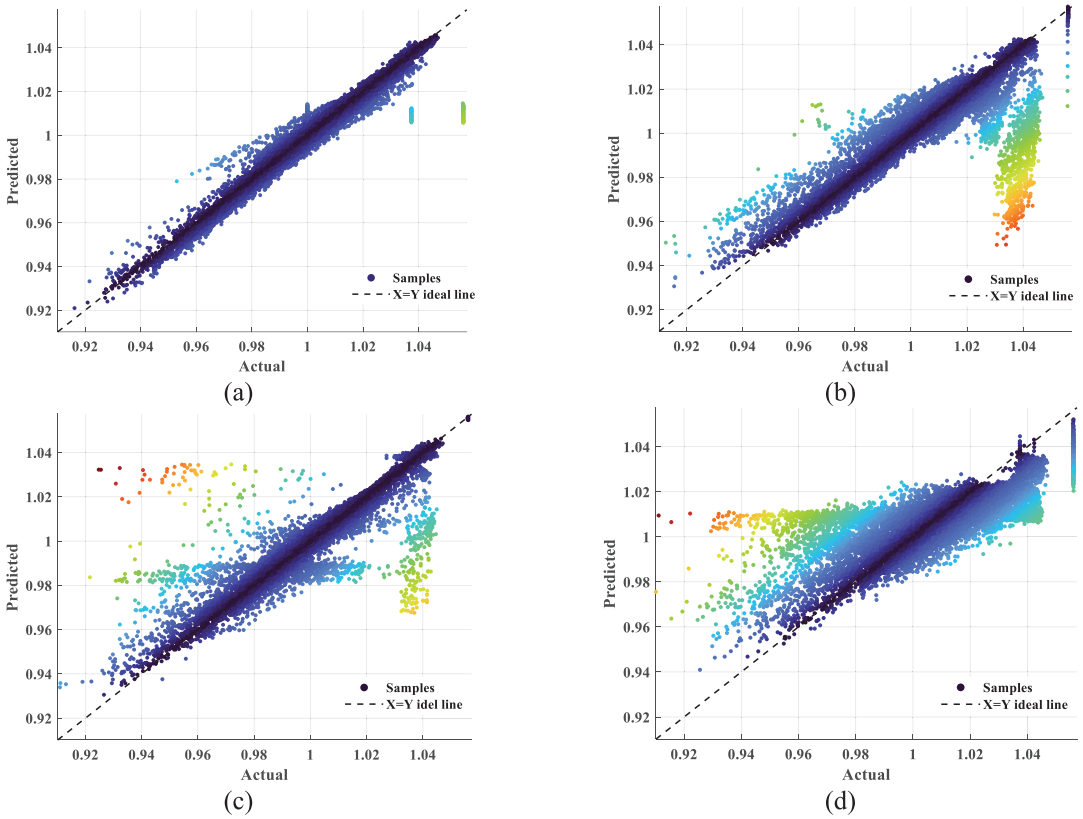


FIGURE 5 | Predicted-actual voltage magnitudes across all test samples for (a) BO-GCN, (b) GCN, (c) GAT and (d) TCN of the IEEE 13-bus system.

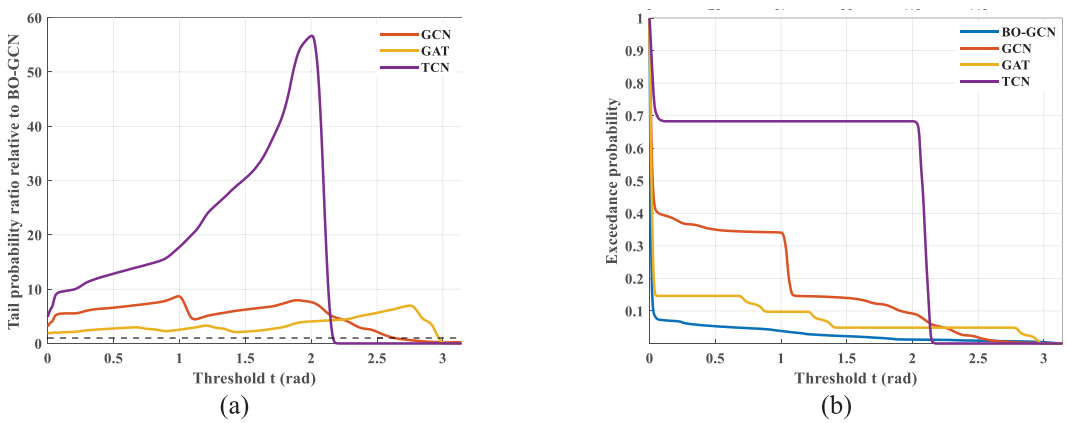


FIGURE 6 | Tail behaviour of absolute phase angle deviation of (a) Tail probability ratio of $P(|\Delta\theta| \geq t)$ for each baseline relative to BO-GCN and (b) Exceedance probability $P(|\Delta\theta| \geq t)$ as a function of threshold t(rad) on the IEEE 13-bus test system.

angle deviations remain several times more frequent than in BO-GCN when operating conditions depart from the dominant patterns. GAT stays closer to the reference at moderate thresholds, but its ratio increases at higher thresholds, indicating occasional fragile behaviour under less common operating combinations even when its bulk behaviour can appear well aligned in scatter space. The most critical behaviour appears in TCN, whose tail ratio grows dramatically around high thresholds, consistent with a structural limitation of non-graph models in representing feeder-dependent sensitivities and phase-coupled responses under changing injections. The CCDF view in Figure 6(b) reinforces the same conclusion by expressing tail risk on a logarithmic probability scale. BO-GCN stays lowest over the full threshold range, meaning that for any chosen tolerance t, the proposed method yields the smallest probability of exceeding that tolerance, which is direct evidence of a lighter tail and stronger robustness. GCN remains above BO-GCN through the mid-range, confirming that a larger fraction of scenarios produces non-trivial angle deviations. GAT also sits above BO-GCN and decays more slowly over higher thresholds, indicating that a non-negligible subset of scenarios still generates large angle errors. TCN shows an abnormal profile with a near-flat exceedance level up to roughly the same high threshold, followed by a sharp drop, which is consistent with systematic large-angle mismatches that

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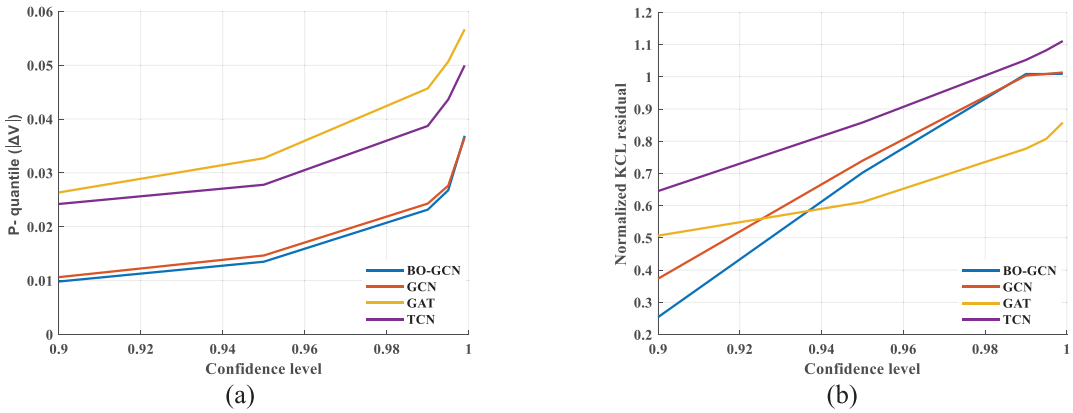


FIGURE 7 | High-confidence performance on the IEEE 123-bus system using (a) p-quantile of voltage-magnitude error and (b) p-quantile of normalised KCL residual.

persist across many operating points and may also reflect wrap handling effects.

4.2 | Case 2: IEEE 123-Bus System

Building on the IEEE 13-bus results, where BO-GCN consistently outperformed the baselines not only in average error metrics but also in tail behaviour, the same models are evaluated on the IEEE 123-bus feeder to test scalability and generalisation under a substantially more challenging unbalanced operating environment. The 123-bus network contains a much larger number of node–phase variables, a richer mix of single, two, and three-phase branches, stronger local phase coupling, and higher diversity in line impedances and loading patterns. These properties amplify nonlinearity and sensitivity in voltage responses, particularly under combined variations of load, PV generation, and EV charging. For this reason, high-confidence performance is emphasised through p-quantile curves, which directly characterise worst-case error levels under increasingly stringent confidence requirements.

In Figure 7(a), all curves increase with the confidence level p, and the steeper rise near the tail, $p \geq 0.99$, indicates that a small subset of scenarios produces disproportionately large voltage errors. BO-GCN remains the lowest curve across the full range, showing the most controlled tail behaviour and the smallest worst-case voltage deviations. This is consistent with its architecture, in which graph convolution aggregates information from electrically connected neighbours, which matches the locality and path-dependent coupling of distribution voltages, while Bayesian hyperparameter tuning improves the capacity–regularisation balance and reduces rare large-error events. The standard GCN is close at moderate p but separates in the tail, suggesting that the inductive bias is correct but the untuned configuration is less robust under rare stressed conditions. GAT exhibits higher quantiles, which can occur in large feeders when attention weights become less stable across heterogeneous neighbourhoods, leading to heavier tail errors. TCN performs worst because it treats node–phase variables as a sequence, and the index ordering does not reflect electrical adjacency, so topology-driven coupling is not learned reliably, especially in extreme scenarios.

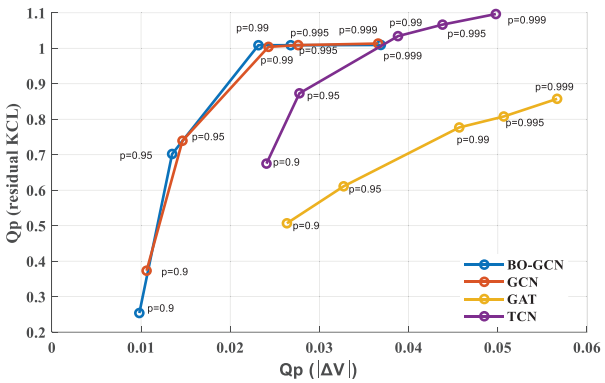


FIGURE 8 | System-level tail trade-off on the IEEE 123-bus system showing paired p-quantiles across confidence levels p.

In Figure 7(b), the KCL residual quantiles increase with p, indicating that the hardest scenarios are also those where physics consistency is most difficult to preserve. BO-GCN achieves the lowest residuals over most confidence levels and remains competitive in the tail, implying more electrically coherent predictions. This follows from message passing, which encourages consistency across connected nodes and therefore reduces current-balance mismatch when evaluated through the network admittance relations. The standard GCN shows a similar trend but degrades more in the tail due to suboptimal tuning. GAT remains between GCN and TCN, reflecting partial locality capture but reduced global coherence under stressed conditions. TCN is consistently highest, indicating systematic physics inconsistency because electrical connectivity is not encoded, so even modest pointwise voltage errors can translate into large KCL mismatches.

To explicitly expose the interaction between voltage-accuracy tails and physics-consistency tails under rare operating conditions, Figure 8 combines both metrics into a single system-level trade-off by plotting, for each model and confidence level p, the paired quantiles $(Q_p(|\Delta V|), Q_p(r_{\mathrm{KCL}}))$.

Across the full range of tail levels, BO-GCN and GCN occupy the left-most region, confirming that they yield the smallest voltage-error quantiles under stressed conditions. For BO-GCN, the tail point at $p = 0.9$ lies near 0.0098 and 0.254 for $Q_{0.90}(|\Delta V|)$ and

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$Q_{0.90}(r_{\mathrm{KCL}})$, respectively, and moves to approximately 0.0135 and 0.702 at $p = 0.95$. The relatively small horizontal displacement indicates that the worst-case voltage deviations remain tightly bounded compared with the other models, while the pronounced upward movement shows that, for extreme scenarios, the dominant degradation manifests as increased KCL mismatch rather than large voltage errors. GCN follows a nearly identical trajectory but is consistently shifted slightly rightward and upward at moderate tails before converging to a similar KCL tail around unity at $p = 0.99$. This separation is consistent with Bayesian hyperparameter tuning improving tail generalisation for voltage prediction while retaining a comparable physics-consistency profile in the most extreme conditions. Moreover, GAT curve is noticeably shifted to the right, indicating larger voltage-error quantiles, yet it remains lower on the vertical axis than the GCN family at the same p, implying better KCL consistency in the tail. Numerically, GAT moves from roughly 0.0264 and 0.507 at $p = 0.90$ to 0.0327 and 0.611 at $p = 0.95$. At high confidence, this value corresponds to substantially smaller KCL tail levels than BO-GCN and GCN, but at the cost of roughly doubling the voltage-error quantile compared with BO-GCN at the same tail level. This pattern suggests that attention-based aggregation may promote more globally coherent predictions with respect to current-balance checks, while introducing a bias or smoothing effect that degrades pointwise voltage accuracy under extreme operating conditions. TCN is dominated by the graph-based approaches and shows the least favourable tail behaviour. Its curve begins at 0.0242 and 0.645 for $p = 0.90$, increases to 0.0278 and 0.858 at $p = 0.95$, and reaches 0.0387 and 1.052 by $p = 0.99$, yielding the highest KCL tail among the compared models at the strictest confidence level reported numerically. The consistent upward and rightward placement is aligned with the architectural limitation of sequence-based models for this task. Without an explicit encoding of electrical adjacency, phase coupling, and topology-driven propagation, tail scenarios tend to amplify both node-level voltage errors and network-level feasibility violations.

To further verify that explicitly encoding network topology is beneficial under highly diverse operating conditions with PV generation and EV charging, the spectral structure of the prediction error is analysed using the Graph Fourier Transform (GFT). This complements the tail-risk and physics-residual results by revealing whether the remaining errors are topologically coherent or fragmented into localised, high-frequency components.

In Figure 9(a), BO-GCN exhibits a dominant low-index component, whereas TCN spreads noticeable energy across a wider set of mid-to-higher graph modes. Because graph-mode index increases with spatial variation on the electrical graph, low-index modes correspond to smooth, feeder-wide deviations, while higher modes correspond to more localized, topology-dependent mismatches. The presence of PV and EV intensifies this distinction. PV injections and EV charging create sharp, spatially clustered power variations and phase-coupled effects that propagate through the feeder impedances. A topology-aware model can partially absorb these effects through neighbourhood aggregation, so its residual error tends to remain topologically structured rather than becoming fragmented across many graph modes. By contrast, a purely data-driven sequence model such as TCN does not encode electrical adjacency or coupling, so scenario-specific local disturbances induced by PV and EV are more likely to appear as scattered spectral components. The GFT cumulative error energy, shown in Figure 9(b), further clarifies this behaviour. The dashed 90% line indicates the graph-mode index at which the cumulative error energy reaches 90% of the total, providing a compact measure of how rapidly the error energy concentrates across the spectrum. Here, TCN reaches high cumulative energy at relatively small mode indices, indicating that a large share of its error energy is captured by the early part of the spectrum, but the curve then saturates quickly and leaves comparatively less gradual accumulation at higher indices. On the other hand, BO-GCN accumulates error energy more gradually across the spectrum. This is consistent with the interpretation that BO-GCN retains a strong structured component but also carries a long, weaker tail over many modes, which is expected when a graph model captures the dominant topology-consistent patterns while residuals remain due to operating-point extreme conditions and localised nonlinearities that vary across PV and EV scenarios.

To support the interpretation of the GFT plots, a compact set of topology-focused spectral descriptors is reported for BO-GCN and TCN in Table 4.

First, the spectral centroid is markedly smaller for BO-GCN than for TCN, meaning the centre of mass of BO-GCN's error spectrum is shifted towards lower graph frequencies. This indicates that the residuals of the topology-aware model remain more topologically coherent and vary more smoothly over the feeder, which is consistent with learning spatial correlations through message passing. A similar conclusion is reinforced by the low-frequency energy ratio, where BO-GCN assigns a larger share of its error energy to the lowest 10% of graph modes than TCN. In practical terms, this suggests that the remaining BO-GCN errors are dominated by global, feeder-wide components that align with impedance-mediated voltage patterns, whereas TCN leaves relatively more energy in higher-frequency content, which is typically associated with localised mismatches around buses with strongly varying injections, such as PV nodes and EV charging points. This contrast becomes clearer when considering the normalised spectral entropy, which is higher for TCN than for BO-GCN. A higher entropy implies that the error energy is spread more uniformly across many graph modes, indicating a less structured and more fragmented residual signature. Under PV- and EV-driven variability, this is an expected outcome for a topology-free model because it lacks an explicit mechanism to align predictions with electrical adjacency and phase-coupled propagation, so local disturbances tend to manifest as spectrally diffuse errors. Finally, the maximum single-mode energy is substantially larger for BO-GCN than for TCN, which is consistent with the obvious low-index peak observed in Figure 9(a). This should not be interpreted as larger absolute error, since the spectrum is normalised; rather, it indicates that the remaining BO-GCN error is concentrated in a small number of coherent graph modes, while TCN distributes its error more evenly across modes. Therefore, when PV generation and EV charging introduce localised and rapidly changing stress patterns, encoding the feeder topology helps confine the residual errors to more graph-smooth, physically interpretable components, whereas ignoring connectivity leads to a more scattered spectral footprint.

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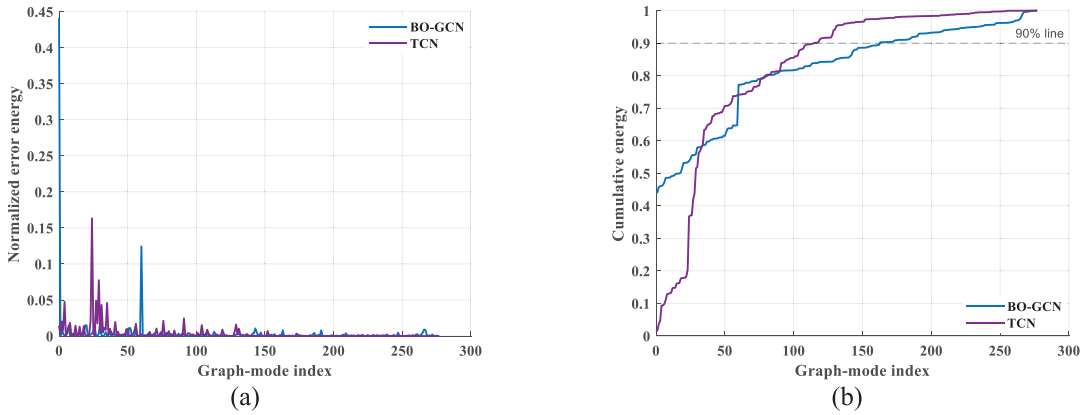


FIGURE 9 | GFT analysis of the prediction error for (a) The normalised error-energy spectrum across graph-mode indices and (b) The corresponding cumulative error-energy curve on the IEEE 123-bus system.

TABLE 4 | Spectral descriptors of the GFT error-energy distribution on the IEEE 123-bus system.

Method	Spectral centroid	Low-frequency energy ratio	Normalised spectral entropy	Maximum single-mode energy
BO-GCN	0.03738	0.55495	0.52852	0.44141
TCN	0.13904	0.37094	0.69227	0.1632

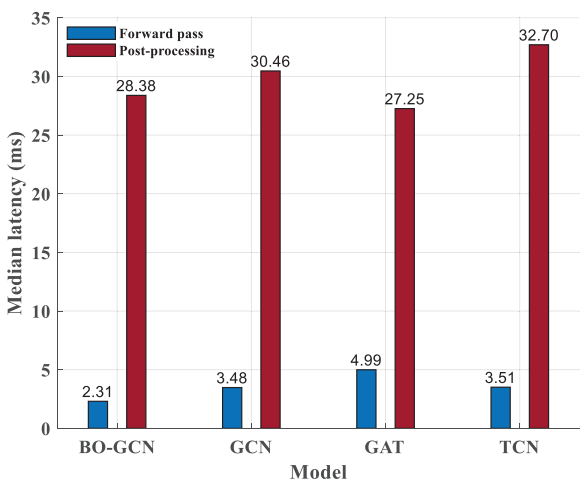


FIGURE 10 | Runtime comparison on the IEEE 123-bus unbalanced system, reporting the median forward-pass inference latency and the median post-processing time.

To further quantify the computational efficiency of the evaluated surrogates, Figure 10 compares the median runtime on the IEEE 123-bus test case by separating the forward-pass latency from the post-processing overhead.

The forward-pass time mainly reflects the computational footprint of the architecture on a fixed graph, scaling with the number of graph layers, hidden width, and edge operations. Accordingly, BO-GCN attains a smaller forward latency because BO selects a more execution-efficient configuration, whereas GAT is slower due to the additional edge-wise attention computations and normalisation, and TCN can remain higher depending on the depth and convolutional operations applied to node-wise sequences. In contrast, the post-processing latency is dominated by evaluation-side computations that are largely model-agnostic, including reconstruction of complex voltages from the $(|V|, \sin\theta, \cos\theta)$ outputs, percentile-based tail statistics, and the KCL residual calculation and current-injection terms. These steps also incur tensor–array conversions and CPU-bound aggregation overhead, which explains why post-processing consistently exceeds the forward pass and exhibits modest variation across models. Finally, BO does not directly increase inference time because it is used offline for hyperparameter selection. Its impact on runtime is indirect and arises only through the chosen hyperparameter configuration that determines the forward-pass cost.

4.3 | Comparative Evaluation With a Literature-Based MLP Baseline

To complement the preceding results on the IEEE 13-bus and IEEE 123-bus feeders, Table 5 reports an additional comparison between the proposed BO-GCN and the Multi-Layer Perceptron (MLP) model simulated from [36]. In [36], the PF problem is formulated as a multi-output regression task, and the MLP is trained on OpenDSS-collected quantities through the OpenDSS–MATLAB COM interface. Following the same general evaluation setting, the MLP baseline was implemented here and tested on the same scenarios and data partitions used for the proposed model, so that the resulting comparison reflects the relative predictive capability of the two surrogates under identical conditions.

Literature-based MLP baseline on both the IEEE 13-bus and IEEE 123-bus feeders. This behaviour is physically expected from the way the two models represent the power-flow mapping.

In the MLP formulation reported in [36], PF prediction is learned as a multi-output regression problem from OpenDSS-collected

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TABLE 5 | Comparative error statistics of the proposed BO-GCN and the MLP baseline on the IEEE 13-bus and IEEE 123-bus systems.

	13-bus				123-bus			
	MAE ($\lvert\Delta V\rvert$)	**MAE ($\lvert\Delta\theta\rvert$)**	**P95 ($\lvert\Delta V\rvert$)**	**P95 ($\lvert\Delta\theta\rvert$)**	**MAE ($\lvert\Delta V\rvert$)**	**MAE ($\lvert\Delta\theta\rvert$)**	**P95 ($\lvert\Delta V\rvert$)**	**P95 ($\lvert\Delta\theta\rvert$)**
BO-GCN	0.0039	0.025	0.0162	0.0569	0.0051	0.0679	0.0141	0.1366
MLP	0.0098	0.1553	0.0248	0.6322	0.0131	0.4184	0.0308	0.7918

quantities, where the input variables are arranged as a single feature vector containing voltage-related quantities, impedance information, and active-reactive load terms. The model then relies on stacked fully connected layers to recover the nonlinear relationship between these global inputs and the target PF outputs. Such a formulation is flexible, but it requires the network to learn feeder interactions, phase coupling, and spatial electrical dependencies implicitly through dense weight matrices. By contrast, the proposed BO-GCN is built directly from the bus–phase representation of the feeder and the nodal admittance matrix. Since each bus-phase node exchanges information through admittance-derived couplings, the model embeds the local electrical interaction pattern of the network during feature propagation rather than leaving it to be inferred only through global dense transformations. This is particularly important in unbalanced feeders, where voltage behaviour is strongly shaped by phase-to-phase coupling, uneven loading, and local topology. As a result, BO-GCN is better aligned with the actual structure of the three-phase PF equations, which explains why it achieves lower average voltage and angle errors in both test systems.

The advantage of BO-GCN is particularly evident in the P95 metrics and becomes more pronounced on the IEEE 123-bus feeder, where the larger dimensionality and stronger structural complexity make the PF mapping more difficult to learn consistently. In the MLP baseline, the entire input-output relationship must be recovered through a single fully connected global mapping, which makes rare or electrically stressed operating points harder to represent, especially for angle prediction and tail behaviour. This difficulty grows with feeder size, as the model must simultaneously capture long-range dependencies, phase asymmetry, and operating-point variability within dense transformations alone. By contrast, BO-GCN remains organised around local electrical neighbourhoods and admittance-weighted interactions, while its bounded voltage decoding and sine–cosine angle representation with unit-circle normalisation improves numerical stability and suppresses unrealistic angular deviations. As a result, the proposed model not only achieves lower average errors but also maintains substantially more controlled voltage and angle tails, particularly in larger and more heterogeneous feeders.

While Table 5 compares the predictive accuracy of BO-GCN and the literature-based MLP baseline, Figures 11(a) and 11(b) provide a complementary physics-orientated view through the Empirical Cumulative Distribution Function (ECDF) of the normalised KCL residual. Because this residual is evaluated from the predicted voltages, injected powers, and feeder admittance relations, it directly reflects how well each model preserves current-balance consistency across the test scenarios.

In both systems, the BO-GCN curve is consistently shifted to the left, indicating that a larger fraction of test samples satisfies the network current-balance relation with smaller residuals. This confirms that the proposed model preserves stronger physics consistency than the MLP baseline, not only in the average sense but across the full error distribution. The gap is especially informative in the tail region, where MLP exhibits a much heavier residual tail, meaning that a non-negligible subset of operating points leads to comparatively large violations of the network equations. This behaviour is physically consistent with the modelling structure of the two approaches. The MLP baseline learns the PF mapping through a fully connected global regression function, so electrical interactions among buses and phases must be inferred only implicitly from the data. By contrast, BO-GCN propagates information over the admittance-informed bus–phase graph, which directly embeds local feeder connectivity and phase coupling during learning. As a result, the predicted voltages remain more compatible with the underlying network equations, yielding smaller KCL residuals. The same pattern is preserved on the IEEE 123-bus system, where the larger size and stronger structural complexity shift both distributions toward higher residuals, but BO-GCN still maintains a substantially more concentrated and physically coherent response than MLP.

Furthermore, in terms of inference cost, the MLP baseline remains computationally lighter, with a median forward-pass latency of approximately 0.4 ms, whereas the proposed BO-GCN requires about 2.31 ms. This difference is expected, since the MLP relies on a simpler fully connected mapping, while BO-GCN performs structure-aware feature propagation over the feeder graph. Nevertheless, the BO-GCN inference time remains in the millisecond range and is still suitable for rapid surrogate evaluation, while providing substantially better prediction accuracy and stronger physics consistency than the MLP baseline.

5 | Conclusion

Three-phase unbalanced distribution PF assessment is increasingly scenario-driven due to high variability in loads and inverter-based resources, yet repeated OpenDSS or nonlinear unbalanced solvers remain computationally intensive for large-scale studies and operational screening. Beyond addressing the computational burden of repeated unbalanced PF evaluation, the main methodological contribution of this work lies in the development of a BO-GCN architecture tailored to three-phase unbalanced distribution feeders. This is achieved through admittance-informed message passing, a stabilised deep graph-convolution encoder, and a physics-guided voltage prediction mechanism that together improve both predictive accuracy and physical plausibility across diverse operating scenarios. This work addressed that scalability challenge by developing a Bayesian-optimised GCN model that predicts per-phase voltage magnitude and angle directly from operating-point injections while

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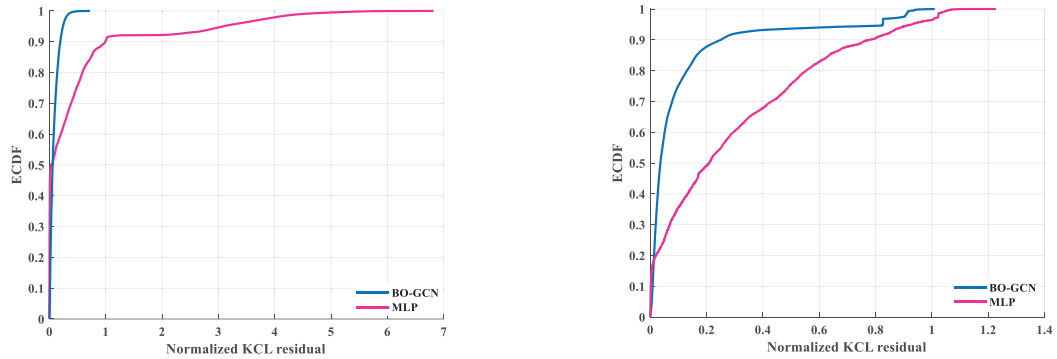


FIGURE 11 | ECDF of the normalised KCL residual for BO-GCN and the MLP baseline on the (a) IEEE 13-bus and (b) IEEE 123-bus systems.

embedding feeder connectivity through admittance-informed message passing and physically structured output heads. Key findings confirm that the proposed BO-GCN provides consistently more accurate voltage-state predictions than baseline data-driven alternatives on standard unbalanced feeders. In the IEEE 13-bus benchmark, BO-GCN reduces voltage-magnitude MAE from 0.0057 to 0.0039 and RMSE from 0.0127 to 0.0094 relative to a standard GCN, corresponding to about 32% and 26% improvements, respectively. The tail voltage error (P95) is also reduced from 0.0291 to 0.0162, yielding about a 44% improvement, indicating stronger reliability near stressed operating conditions. Similar accuracy and stability trends are observed across larger test feeders, demonstrating that admittance-weighted aggregation and bounded voltage-angle decoding improve generalisation under diverse DER and EV scenarios. Results on the larger feeder validate scalability, and the P95–P99 analysis provides evidence that accuracy does not degrade disproportionately in extreme cases. These results suggest that electrically informed graph surrogates can serve as practical fast-evaluation tools for time-series studies, uncertainty analysis, and planning loops that require thousands of unbalanced operating points while maintaining improved physical plausibility compared with purely vectorised predictors.

Nomenclature

Set and Indices

$\mathcal{G} = (\mathcal{N}, \mathcal{M})$	Unbalanced distribution network
$\mathcal{N}$	Set of buses
$\mathcal{M}$	Set of lines
Γ	Set of phases
$\Gamma_i \subseteq \Gamma$	Set of phases available at bus i
$\phi, \varphi \in \Gamma$	phase indices
s	Scenarios index
N_s	Number of scenarios
n	Number of bus–phase nodes
$\mathcal{K}$	Set of bus–phase nodes

Variables and Parameters

$V_i^\phi \in \mathbb{C}$	Complex voltage at bus i, phase ϕ
$\lvert V_i^\phi \rvert$	Voltage magnitude at bus i, phase ϕ
θ_i^ϕ	Voltage angle at bus i, phase ϕ
$I_{ij}^\phi \in \mathbb{C}$	Complex current on branch (i,j), phase ϕ
$S_i^\phi \in \mathbb{C}$	Complex power injection at bus i, phase ϕ
P_i^ϕ, Q_i^ϕ	Active and reactive power injections
$S_i^{\phi,(s)}, P_i^{\phi,(s)}, Q_i^{\phi,(s)}$	Scenario-wise constant-PQ injections
$Z_{ij} \in \mathbb{C}^{\lvert\Gamma_{ij}\rvert \times \lvert\Gamma_{ij}\rvert}$	Phase-coupled impedance matrix of branch (i, j)
$Y \in \mathbb{C}^{n \times n}$	Bus–phase nodal admittance matrix
$diag(\cdot)$	Diagonal matrix operator
$f_\Theta(\cdot)$	Learned surrogate mapping with parameters Θ
Θ	Trainable parameters of the learning model
$x^{(s)}$	Input feature matrix for scenario s
$y^{(s)}$	Target matrix for scenario s
$\mu_P, \sigma_P, \mu_Q, \sigma_Q$	Training-set mean and standard deviation for standardization

Author Contributions

Amir Hossein Baharvand: data curation, formal analysis, methodology, software and writing – original draft. **Meysam Doostizadeh:** conceptualisation, supervision, validation, visualisation, writing – review and editing, **Mahmoudreza Shakarami:** investigation, resources, supervision, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets generated and analysed during this study are available from the corresponding author upon reasonable request.

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